

Electromagnetism

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Q1. When a coil is placed in a time dependent magnetic field the power dissipated in it is P . The number of turns, area of the coil and radius of the coil wire are N , A and r respectively. For a second coil with number of turns, area and radius $2N$, $2A$ and $3r$ respectively, the power dissipated is $\sqrt{2} \alpha P$. Find α .

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- (1) 36
- (2) $128\sqrt{2}$
- (3) 16
- (4) 64

✓ **Answer: (1) 36**

Solution:

EMF = $N \times A \times (dB/dt)$ (unchanged by wire radius).

Resistance depends on wire length and its own cross-sectional area πr^2 , giving $P \propto N \times A^{1.5} \times r^2$ (derived from $R = \rho \times \text{length} / (\pi r^2)$ with length $\propto N \times \sqrt{A}$).

For the second coil: $P' \propto 2N \times (2A)^{1.5} \times (3r)^2 = 2 \times 2^{1.5} \times 9 \times NA^{1.5}r^2 \approx 50.9 \times (NA^{1.5}r^2)$.

$P'/P \approx 50.9$. Since $P' = \sqrt{2} \alpha P$, $\alpha = 50.9/\sqrt{2} \approx 36$.

Q2. An electromagnetic wave travelling in x-direction is described by $E_y = 300 \sin \omega(t - x/c)$. If an electron is restricted to move in the y-direction only with speed 1.5×10^6 m/s, the ratio of maximum electric and magnetic forces acting on the electron is ____.

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- (1) 200
- (2) 150
- (3) 400
- (4) 300

✓ **Answer: (1) 200**

Solution:

Max $E_0 = 300$ V/m, so max $B_0 = E_0/c = 300/(3 \times 10^8) = 1 \times 10^{-6}$ T.

Electric force $F_E = eE$. Magnetic force $F_B = evB$ (v perpendicular to B , along y , B along z).

Ratio $F_E/F_B = E/(vB) = 300/(1.5 \times 10^6 \times 1 \times 10^{-6}) = 300/1.5 = 200$.